

House Price Predictor

California Housing • Linear Regression

- Linear Regression on the California Housing dataset
- End-to-end machine-learning portfolio project

Problem and data

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- Predict median house value (MedHouseVal)
- 20,640 California block groups and 8 numeric features
- Target unit: \$100,000s of 1990 USD

Workflow

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- Load and explore data
- Coerce invalid/blank entries; median-impute missing values
- 80/20 train-test split with `random_state=42`
- Train a `LinearRegression` pipeline

Exploration

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- Summary statistics and missing-value counts
- Feature and target distributions
- Correlation with price and geography
- Diagnostic plots: actual vs predicted and residuals

Held-out evaluation

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- MAE: 0.5332 \$100k
- RMSE: 0.7456 \$100k
- R^2 : 0.5758
- Metrics calculated on 4,128 untouched test rows

Outputs

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- Saved preprocessing + model pipeline (.joblib)
- Metrics, predictions, EDA tables, and report
- Notebook plus editable Marp slide source

Next steps

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- Compare Ridge/Lasso and tree-based models
- Use cross-validation and current property-level data
- Evaluate bias, drift, and local error before deployment